

Quiz 1

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (7 points) Given $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (1, 5)$.

(a) Find the *cosine* of the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{1^2 + 2^2} = \sqrt{1+4} = \sqrt{5}$$

$$\|\mathbf{w}\| = \sqrt{1^2 + 5^2} = \sqrt{1+25} = \sqrt{26}$$

$$\mathbf{v} \cdot \mathbf{w} = 1 \times 1 + 2 \times 5 = 11$$

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{11}{\sqrt{5} \cdot \sqrt{26}} = \frac{11}{\sqrt{130}}$$

(b) Choose a number c so that $\mathbf{w} + c\mathbf{v}$ is perpendicular to $\mathbf{v}$. (*Hint:* using $(\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$)

$$(\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$$

$$\mathbf{w} \cdot \mathbf{v} + c\mathbf{v} \cdot \mathbf{v} = 0$$

$$c = -\frac{\mathbf{w} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}$$

$$= -\frac{11}{5}$$

Problem 2. (3 points) Compute $\mathbf{Ax}$ (i.e., the matrix $\mathbf{A}$ multiplies the vector $\mathbf{x}$) with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 0 & -1 \\ -4 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

by using linear combination of the columns of $\mathbf{A}$ with components of $\mathbf{x}$ as coefficients.

$$\begin{aligned} \mathbf{Ax} &= 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -4 \\ -8 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 2+4+0 \\ -4+0-3 \\ -8+2+6 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ -7 \\ 0 \end{bmatrix} \end{aligned}$$